STEGANOGRAPHY VS CRYPTOGRAPHY: HIDDEN SECURITY IN THE DIGITAL WORLD

CSA 5111- CRYPTOGRAPHY AND NETWORK SECURITY

Technical Presentation

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INTRODUCTION

- In today's digital world, large amounts of information are shared through the internet, making the protection of sensitive data increasingly important.
- During digital communication, confidential information may be exposed to cyberattacks, unauthorized access, or misuse if proper security measures are not followed.
- To address these challenges, various information security techniques are used to protect data and ensure safe communication between individuals and organizations.
- Steganography and cryptography are two important techniques that protect secret information in different ways. Steganography hides the existence of a message, whereas cryptography converts the message into an unreadable format.
- This presentation explores the working principles, types, applications, advantages, and limitations of both techniques, along with a comparison of how they contribute to information security.

STEGANOGRAPHY — DEFINITION AND WORKING

- **Steganography** is the technique of hiding secret information inside an ordinary file, such as an image, audio, video, or text file, so that the existence of the secret message is not noticed.

Working of Steganography

1. **Secret Message:** The sender selects the message or information to be hidden.
2. **Cover File:** An ordinary image, audio, video, or text file is selected to hide the message.
3. **Embedding:** The secret message is hidden inside the cover file using a steganography technique.
4. **Stego File:** The resulting file contains the hidden message and looks almost identical to the original file.
5. **Transmission:** The stego file is sent to the receiver.
6. **Extraction:** The receiver uses the appropriate technique to extract the hidden message.

TYPES AND APPLICATIONS OF STEGANOGRAPHY

Types of Steganography

Steganography can be classified based on the type of cover medium used to hide the secret message.

1. Image Steganography



- Hides data in image files using techniques like LSB (Least Significant Bit), pixel modification, or DCT coefficients.
- Common formats: BMP, PNG, JPEG.

2. Audio Steganography



- Hides data in audio files by modifying audio samples (e.g., LSB, phase coding, echo hiding).
- Common formats: WAV, MP3, AAC.

3. Video Steganography



- Hides data in video frames using techniques like frame modification, frequency domain, or temporal coding.
- Common formats: MP4, AVI, MOV.

4. Text Steganography



- Hides data in text using techniques like white space manipulation, word substitution, or formatting.
- Common formats: TXT, HTML, PDF.

5. File / Format Steganography



- Hides data in other file types (e.g., executable files, PDF, DOCX) by using unused space or metadata.
- Common formats: EXE, PDF, DOCX, ZIP.

Applications of Steganography

Steganography is widely used in areas where secret communication and data hiding are required.

1. Secure Communication

- Hides secret messages in ordinary files (images, audio, etc.).
- Used by individuals, journalists, and organisations for private communication.

2. Digital Watermarking

- Embeds ownership or copyright information in media files.
- Used in content protection, media distribution and anti-piracy.

3. Authentication & Integrity

- Hides a verification code or hash inside a file.
- Helps in proving the authenticity of digital content.

4. Covert Data Storage

- Used to store small amounts of data secretly within larger files.
- Useful for bypassing restrictions or censorship.

5. Cybersecurity & Penetration Testing

- Used for hiding malware commands or exfiltrating data.
- Helps attackers avoid detection by security systems.

6. Military and Government

- Used for classified communication and intelligence gathering.
- Helps in covert operations and secure information exchange.

7. Forensic and Law Enforcement

- Used to hide or detect hidden data in digital evidence.
- Helps in investigations and digital forensics.

PRACTICAL DEMONSTRATION

Caesar Cipher

Problem

Encrypt the plaintext **CAT** using a Caesar cipher with key $K = 3$.

Algorithm:

Encryption formula:

$$C = (P + K)\ 26$$

Use the letter values A = 0, B = 1, ..., Z = 25.

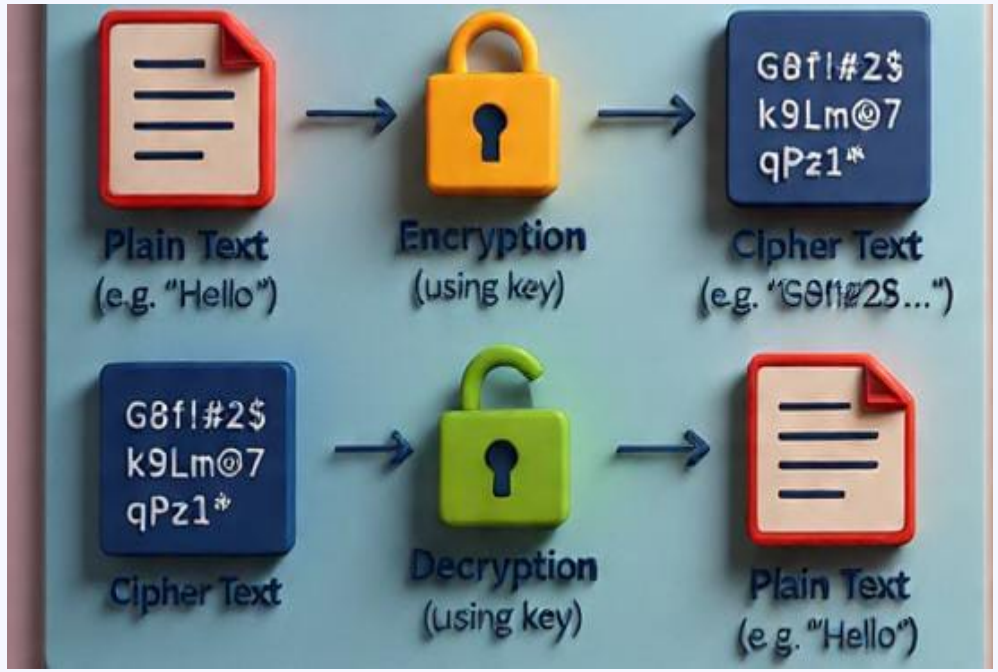
Manual Calculation

Plaintext	Value (P)	Calculation	Ciphertext
C	2	(2 + 3) mod 26 = 5	F
A	0	(0 + 3) mod 26 = 3	D
T	19	(19 + 3) mod 26 = 22	W

Answer: CAT → FDW

CRYPTOGRAPHY — DEFINITION AND WORKING

- **Cryptography** is the technique of protecting information by converting readable data (**plaintext**) into an unreadable form (**ciphertext**) using an encryption algorithm and a key.



Working of Cryptography

1. **Plaintext:** The sender enters the original message.
2. **Encryption:** The message is converted into an unreadable format using an algorithm and a key.
3. **Ciphertext:** The encrypted message is sent to the receiver.
4. **Decryption:** The receiver uses the appropriate key to convert the ciphertext back into the original message.
5. **Original Message:** The receiver reads the decrypted message

Types and Applications of Cryptography

Types of Cryptography

Cryptography can be classified into different types based on the kind of operations used, the number of keys, and the purpose it serves.

1. Symmetric Key Cryptography
(Secret Key Cryptography)



- Uses the same key for both encryption and decryption.
- Fast and efficient.
- **Examples:** AES, DES, 3DES, ChaCha20.
- Best for bulk data encryption.

2. Asymmetric Key Cryptography
(Public Key Cryptography)



- Uses a pair of keys: public and private key.
- Slower than symmetric key cryptography.
- **Examples:** RSA, ECC, Diffie-Hellman.
- Used for secure key exchange, digital signatures.

3. Hash Function Cryptography
(One-way Hashing)



- Converts data into a fixed-length string (hash).
- One-way (cannot be reversed).
- Used for data integrity and authentication.
- **Examples:** SHA-256, SHA-3, MD5 (legacy).

4. Hybrid Cryptography
(Combination of Symmetric + Asymmetric)



- Uses symmetric encryption for data and asymmetric encryption to exchange the session key.
- Combines speed and security.
- **Examples:** TLS/SSL, PGP.
- Widely used in secure communication.

5. Stream Cipher



- Encrypts data bit by bit (or byte by byte).
- Uses a pseudo-random keystream.
- **Examples:** RC4, Salsa20, ChaCha20.
- Suitable for real-time communication.

6. Block Cipher



- Encrypts data in fixed-size blocks.
- Uses modes like CBC, ECB, OFB, CTR.
- **Examples:** AES, DES, Blowfish.
- Commonly used for file and disk encryption.

7. Public Key Infrastructure (PKI)



- Manages digital certificates and key pairs.
- Provides authentication, integrity and non-repudiation.
- **Examples:** X.509 certificates, Certificate Authority (CA).

Applications of Cryptography

Cryptography is widely used in various fields to ensure security, privacy and trust in digital and real-world systems.

1. Secure Communication



- Protects messages in emails, instant messaging and VoIP.
- Used in HTTPS, VPNs and secure chat apps.
- **Example:** TLS/SSL.

2. Data Protection and Storage



- Encrypts sensitive data at rest and in transit.
- Used in cloud storage, hard drives and databases.
- **Example:** AES, BitLocker.

3. Digital Signatures



- Verifies the authenticity of the sender.
- Ensures data integrity and non-repudiation.
- Example: RSA, ECDSA.

4. E-commerce and Online Transactions



- Secures online payments and financial transactions.
- Protects credit card details and user identity.
- **Example:** SSL/TLS, SET.

5. Authentication and Access Control



- Verifies user identity.
- Used in login systems, biometrics and multi-factor authentication.
- **Example:** Password hashing, Kerberos.

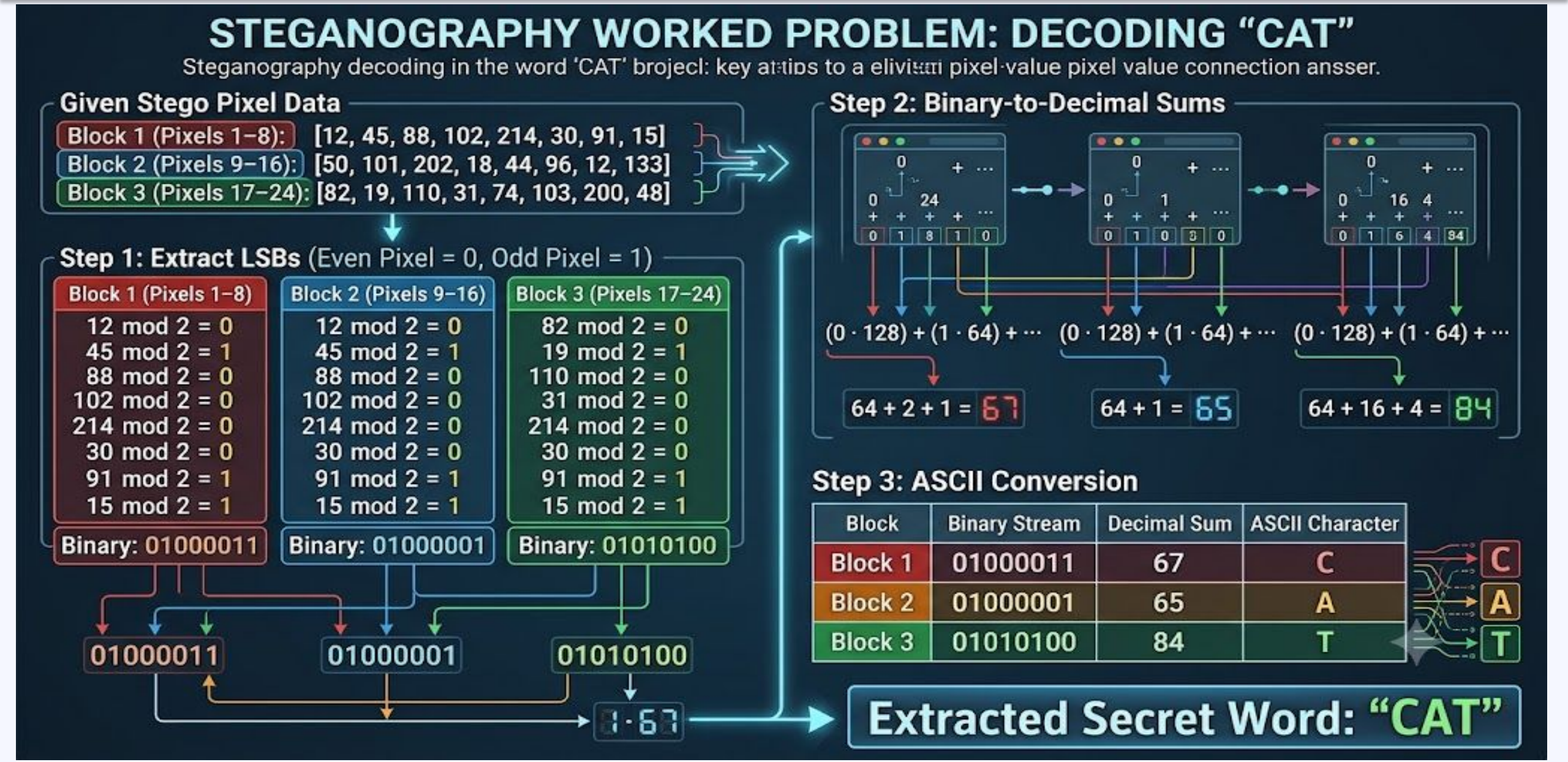
6. Government and National Security



- Protects classified information and critical infrastructure.
- Used in e-governance, digital identity and defense systems.
- **Example:** PKI, encryption standards.

In short: Cryptography protects data, ensures privacy, builds trust and enables secure communication across digital and real-world systems.

PRACTICAL DEMONSTRATION



STEGNOGRAPHY VS CRYPTOGRAPHY

S.No.	Cryptography	Steganography
1.	Converts a message into an unreadable form.	Hides a secret message inside another file.
2.	Protects the meaning of the message.	Hides the existence of the message.
3.	The encrypted message may look suspicious.	The file usually looks normal.
4.	Uses encryption and decryption algorithms.	Uses embedding and extraction techniques.
5.	Requires a key for encryption and/or decryption, depending on the method.	May use a key to hide and retrieve the secret message.
6.	The output is called ciphertext.	The output is called a stego-file.
7.	Example: Converting HELLO into KHOOR.	Example: Hiding HELLO inside an image.
8.	Used in secure messaging, banking, and data protection.	Used in secret communication and digital watermarking.

Thank You